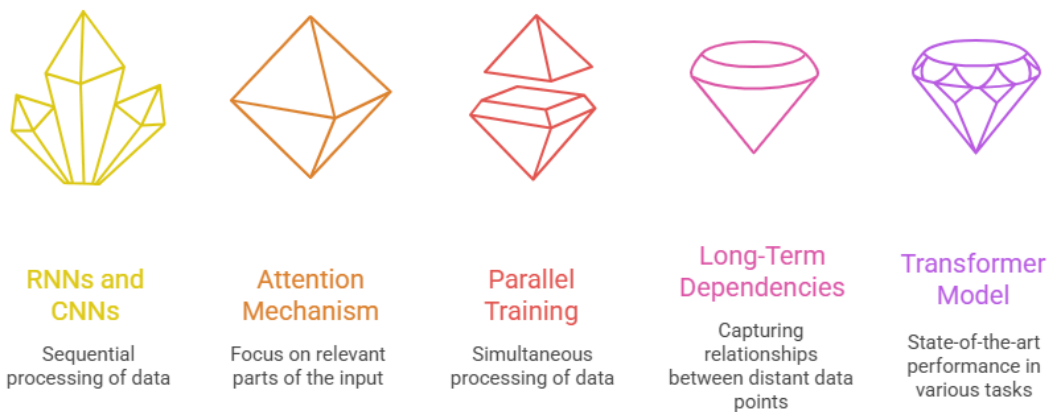


Transformers, BERT, and GPT: The Backbone of Modern Deep Learning

Overview

The Transformer architecture revolutionized deep learning by replacing RNNs and CNNs with a model built entirely on the Attention Mechanism.

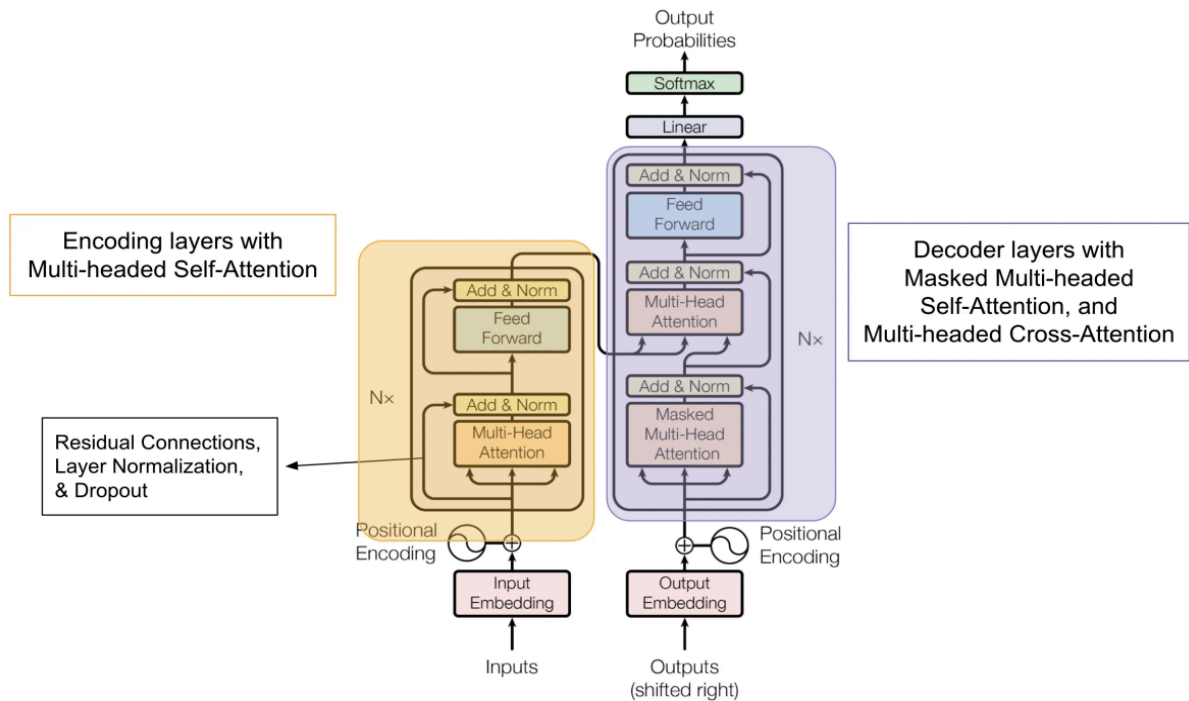
- Proposed by Vaswani et al. (2017) in the paper “Attention is All You Need.”
- Enabled parallel training, better handling of long-term dependencies, and state-of-the-art performance in NLP, vision, and multimodal tasks.



Transformers are the foundation of BERT, GPT, T5, Vision Transformers (ViT), and modern generative AI systems like ChatGPT.

Transformer Architecture

The Transformer consists of an Encoder-Decoder structure.



◆ Encoder

- Takes input sequence.
- Each layer has:
 1. Multi-Head Self-Attention → each token attends to all others.
 2. Feed-Forward Network (FFN) → nonlinear transformations.
 3. Residual Connections + Layer Normalization → stability.

◆ Decoder

- Generates output sequence (e.g., translation).
- Each layer has:

1. Masked Multi-Head Attention → prevents looking at future tokens.
2. Encoder-Decoder Attention → attends to encoder outputs.
3. Feed-Forward + Residual + Normalization.

♦ Positional Encoding

- Since transformers don't use recurrence or convolution, they add positional encoding (sine & cosine functions) to represent word order.

Formula for self-attention:

$\text{Attention}(Q,K,V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$

Why Transformers Beat RNNs/LSTMs

- Parallelization → process all tokens at once (vs sequential RNNs).
- Handles long dependencies → no vanishing gradient issues.
- Scalable → trains on massive datasets efficiently.
- Universal → works in NLP, Vision, Speech, Multimodal AI.

BERT (Bidirectional Encoder Representations from Transformers)

BERT (Google, 2018) is a Transformer-based encoder-only model.

♦ Key Features

- **Bidirectional:** Looks at context from both left and right.
- **Pretraining Objectives:**
 - Masked Language Modeling (MLM) → Randomly mask words and predict them.
 - Next Sentence Prediction (NSP) → Predict if sentence B follows A.

- Transfer Learning: Pretrained on large corpus → fine-tuned for tasks.

♦ Applications

- Question answering (SQuAD benchmark).
- Sentiment analysis.
- Named entity recognition (NER).
- Text classification.

Example:

Sentence: The cat sat on the [MASK].

BERT predicts: mat.

GPT (Generative Pretrained Transformer)

GPT (OpenAI, 2018 → GPT-1, GPT-2, GPT-3, GPT-4) is a decoder-only Transformer model.

♦ Key Features

- Autoregressive: Predicts the next word given previous ones.
- Pretraining Objective:
 - Causal Language Modeling → left-to-right prediction.
- Scaling Laws: Larger models + more data → better performance.
- Few-shot & Zero-shot Learning: GPT-3 showed the ability to perform tasks without fine-tuning.

◆ Applications

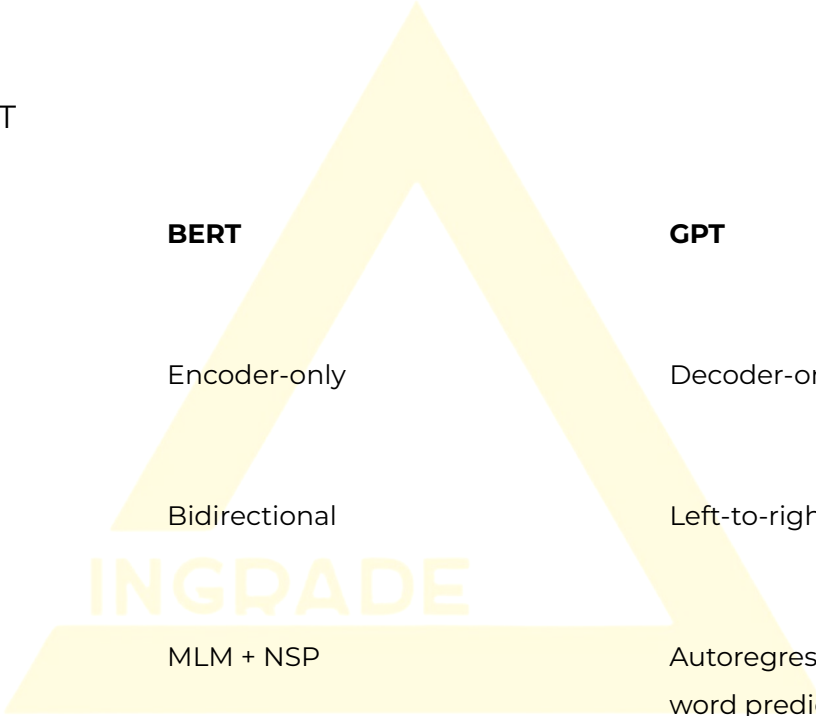
- Text generation (ChatGPT).
- Code generation (Codex).
- Summarization.
- Dialogue systems.

Example:

Input: "Once upon a time"

GPT generates: "... there was a king who ruled a vast kingdom."

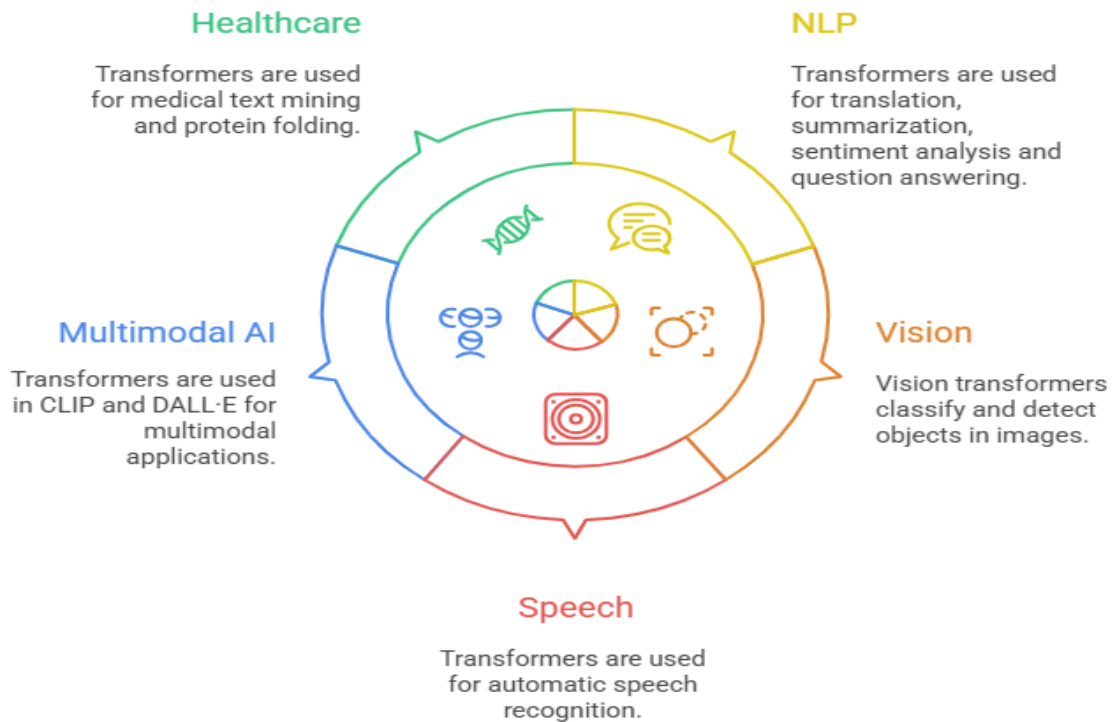
BERT vs GPT



Aspect	BERT	GPT
Architecture	Encoder-only	Decoder-only
Context	Bidirectional	Left-to-right (causal)
Objective	MLM + NSP	Autoregressive (next word prediction)
Usage	Understanding tasks (classification, QA)	Generation tasks (text, code, stories)
Example Models	BERT, RoBERTa, ALBERT	GPT-2, GPT-3, GPT-4, ChatGPT

Applications of Transformers

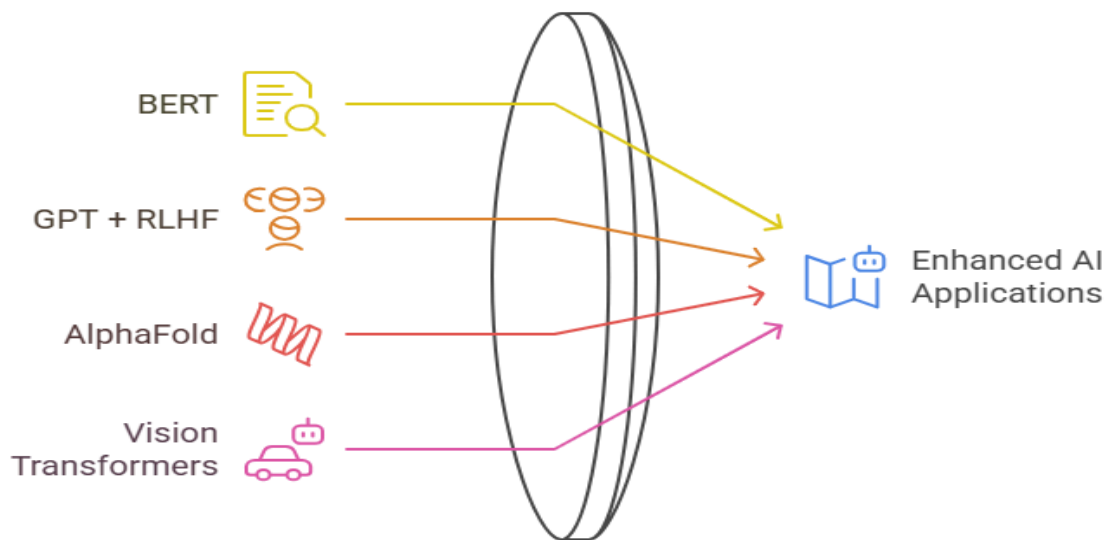
Transformer Applications



- NLP → Translation, summarization, sentiment analysis, QA.
- Vision → Vision Transformers (ViT) classify & detect images.
- Speech → ASR (Automatic Speech Recognition).
- Multimodal AI → CLIP (text + image), DALL-E (text → image).
- Healthcare → Medical text mining, protein folding (AlphaFold).

Real-World Impact

AI's Transformative Impact



- Google Search → Uses BERT for query understanding.
- ChatGPT → Built on GPT + Reinforcement Learning from Human Feedback (RLHF).
- DeepMind AlphaFold → Transformer-based model solved protein folding.
- Autonomous Vehicles → Vision Transformers for object detection.

Challenges of Transformers

- High computational cost → needs GPUs/TPUs.
- Data hungry → requires huge datasets.
- Interpretability → hard to understand why they work.
- Bias → may learn social and cultural biases from data.

Key Takeaways

- Transformers replaced RNNs as the new foundation of AI.
- BERT = Encoder-only → great for understanding tasks.
- GPT = Decoder-only → great for generation tasks.
- Modern AI systems (ChatGPT, BERT-based search, AlphaFold) all rely on Transformer architectures.

